ANALYSIS OF NEW YORK CITY UBER BOOKING PROCEEDS

UTILIZING ALTAIR AI STUDIO - RAPIDMINER

Applied Machine Learning Project

Kiruthika Muralidharan

MOTIVATION:

Ride-sharing platforms like Uber face an operational challenge in optimally pricing ride bookings for maximum profit and business stability. This pricing includes considerations for not only Uber's cut of proceeds but also for the financial welfare of their drivers, whom their business model depends upon.

OBJECTIVE:

This project focuses on gaining an understanding of the pricing model currently being utilized by Uber in New York City by analyzing correlations between collected data points such as trip miles/trip time/tolls/fees/surcharges and income such as driver pay/Uber compensation.

In addition to this financial modeling, cluster analysis will create optimal customer segmentation with descriptive statistics for each segment that can inform future business decisions concerning pricing, marketing, and negotiations with both drivers and external parties.

KEY RESEARCH QUESTIONS:

- What variables have the largest effect on ride proceeds (Uber pay/driver pay)?
- Can we predict the ride proceeds (Uber pay/driver pay) using trip information provided by New York City's government (or drivers themselves)?
- How can Uber best describe the ride-booking segments that exist in New York City?
- What insights can be gained by reviewing descriptive statistics of these ride-booking segments?
- How can these insights be applied to improve operational decision-making?

POTENTIAL APPLICATIONS:

- Optimizing pricing algorithms and policies
- Designing driver retention and engagement strategies
- Leveraging data-driven insights in negotiations with drivers, local transit systems, and governing bodies
- Improving marketing strategies through booking segment insights

DATASET DESCRIPTION:

The analysis uses a sample of a **secondary dataset** derived from September 2025 high-volume, for-hire-vehicle bookings publicly available at nyc.gov:
https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

The dataset contains a **random sampling of 100,000 Uber bookings from September 2025 in CSV format**, providing detailed records of trip pick-up/drop-off locations, distances, fares, tolls, fees, and behavioral flags. These data points provide valuable information regarding operational patterns and pricing dynamics.

TARGET VARIABLES:

uber_pay – This is a continuous numerical variable representing the amount of money paid to Uber for each ride-share booking and is derived from **base_passenger_fare minus driver_pay**.

driver_pay — This is a continuous numerical variable representing the amount of money paid to the driver for each ride-share booking.

DATASET DESCRIPTIVE STATISTICS:

To support the project's objective of understanding Uber's pricing structure and income distribution in New York City, descriptive statistics were generated for the 100,000 randomly sampled Uber trips from September 2025. These statistics provide essential context for the later correlation analysis, predictive modeling, and clustering work completed in this study.

Trip Characteristics:

The data shows that the majority of Uber rides in New York City are short-distance, short-duration urban trips. The average trip distance is 5.26 miles (median 3.11 miles), while the maximum reaches 139 miles. Trip times show a similar pattern, averaging 1,270 seconds (~21 minutes) with a median of 1,028 seconds.

Both variables are heavily right-skewed, confirming that while most trips are short and routine, a small number of very long-distance rides significantly increase the upper range of both miles and time. This skewness helps explain many patterns observed later in the financial modeling, particularly the variability seen in Uber's revenue.

Fare Components and Surcharges:

The fare system in NYC consists of base fare plus tolls, taxes, and regulated fees. The mean base passenger fare is $29.43, with a median of $21.09, again indicating a highly skewed distribution. Tolls, sales tax, Black Car Fund (bcf), and congestion surcharge contribute smaller but consistent amounts. Airport fees and certain zone-based surcharges apply only to specific trip types, resulting in lower overall averages.

These patterns reflect the complexity of NYC's pricing structure, where regulatory rules, congestion zones, airport areas, and mandatory state fees all contribute to the total fare. Understanding these components is important for interpreting Uber's and drivers' earnings in later analyses.

Driver Pay, Uber Pay, and Earnings Rates:

The core financial variables—driver_pay, uber_pay, and tips—show substantial heterogeneity, which aligns with the research questions focusing on pricing, prediction, and segmentation. Driver pay averages $22.56 (median $16.88), representing the more stable portion of the payout structure. Uber's commission (uber_pay) averages $6.86, but ranges from –$64 (subsidized or regulated trips) to $196 (premium long rides).

On a per-second earnings basis:
Drivers earn ~1.8¢/sec
Uber earns ~0.7¢/sec, but with greater volatility

This variability becomes highly relevant in the clustering analysis, where long rides and MTA/on-behalf-of rides form distinct segments largely based on these financial differences.

Tips:

Tips show strong zero inflation, consistent with typical ride-share tipping behavior. The average tip is $1.27, median is $0, and maximum is $98. Since most riders do not tip at all, this variable contributes highly unevenly to driver earnings.

Behavioral and Accessibility Flags:

Several binominal flags in the dataset identify special ride types: Shared rides occur in 2–3% of the sample. Wheelchair-accessible (WAV) requests and matches are rare (<1%). MTA Access-A-Ride bookings represent only ~0.1% of rides. Despite their low frequency, these ride types often involve subsidies or regulatory pricing rules. This explains why negative Uber pay values appear in the dataset and why certain clusters later emerge as outliers.

Overall:

Overall, Uber's September 2025 NYC trip data reveals a highly predictable and structured marketplace. The findings provide immediate, data-backed opportunities for pricing improvements, driver engagement, marketing efficiency, and regulatory planning.

Descriptive Statistics Table:

Attribute	Type	Min/Least	Max/Most	Mean	Median	Std Dev	Sum	Description
PULocationID	nominal	99 (1 rows)	138 (2051 rows)	NA	NA	NA	NA	Taxi Zone of pick-up
DOLocationID	nominal	99 (2 rows)	265 (5013 rows)	NA	NA	NA	NA	Taxi Zone of drop-off
trip_miles	numerical	0	139.81	5.261	3.11	6.134	526132.78	Total miles of trip
trip_time	numerical	45	13676	1270.023	1028	917.346	127002271	Total time (seconds) of trip
base_passenger_fare	numerical	-1.92	590.47	29.426	21.09	27.115	2942582.03	Fair before tolls, tips, taxes, and fees
tolls	numerical	0	52.44	1.215	0	3.713	121482.04	Total of trip tolls
bcf	numerical	0	15.67	0.730	0.52	0.686	73012.56	Total collected for Black Car Fund in trip
sales_tax	numerical	0	40.42	2.459	1.79	2.224	245903.98	Total collected for NYS sales tax in trip
congestion_surcharge	numerical	0	2.75	1.042	0	1.328	104182.00	Total collected for NYS congestion surcharge in trip
airport_fee	numerical	0	5.00	0.226	0	0.719	22572.50	Total airport fee in trip
tips	numerical	0	98.00	1.269	0	3.849	126850.66	Total driver tips
driver_pay	numerical	0	464.99	22.562	16.88	19.605	2256246.29	Total driver pay before tips, commission, surcharges, or taxes
shared_request_flag	binominal	Y (3619 rows)	N (96381 rows)	NA	NA	NA	NA	Agreed to shared ride (Y/N)
shared_match_flag	binominal	Y (2096 rows)	N (97904 rows)	NA	NA	NA	NA	Actual shared ride (Y/N)
access_a_ride_flag	binominal	Y (114 rows)	N (99886 rows)	NA	NA	NA	NA	On behalf of MTA (Y/N)
wav_request_flag	binominal	Y (304 rows)	N (99696 rows)	NA	NA	NA	NA	Requested wheelchair-accessible vehicle (Y/N)
wav_match_flag	binominal	Y (9772 rows)	N (90228 rows)	NA	NA	NA	NA	Actual wheelchair-accessible vehicle (Y/N)
cbd_congestion_fee	numerical	0	1.50	0.549	0	0.723	54940.50	MTA's Congestion Relief Zone charge
Derived Atrribute	**Type**	**Min/Least**	**Max/Most**	**Mean**	**Median**	**Std Dev**	**Sum**	**Description**
uber_pay	numerical	-64.02	196.06	6.863	4.43	11.570	686335.74	base_passenger_fare - driver_pay
uber_pay_rate	numerical	-0.097	0.256	0.007	0.006	0.008	NA	(base_passenger_fare - driver_pay) / trip_time
driver_pay_rate	numerical	0	0.311	0.018	0.016	0.006	NA	driver_pay / trip_time

ANALYTICAL GOAL AND MODELING:

The goal is to develop a supervised, machine-learning model capable of predicting the numerical outcomes of driver_pay and uber_pay with a high degree of accuracy. Our selected models include **linear regression** and **neural networks**. Creating a successful prediction model will reveal the direction and strength of relationships between the dependent income attributes and other variables.

Our secondary goal is to develop an unsupervised, machine-learning model capable of predicting optimal ride-booking segments using **K-Means Clustering**. Descriptive statistics of these segments will provide insights into customer and driver behaviors along with the real work effects of the current pricing model.

CORRELATIONS AND ATTRIBUTE SELECTION:

A correlation matrix is included below. To be a useful model, certain attributes should be excluded from the model. uber_pay, driver_pay, sales_tax, and base_passenger_fare are all attributes determined by the pricing model directly or indirectly and should be excluded from the model. In addition, ride flag correlations are low and would not benefit the model. congestion_surcharge and cbd_congestion_fee are strongly correlated, so only one attribute needs to be included between them.

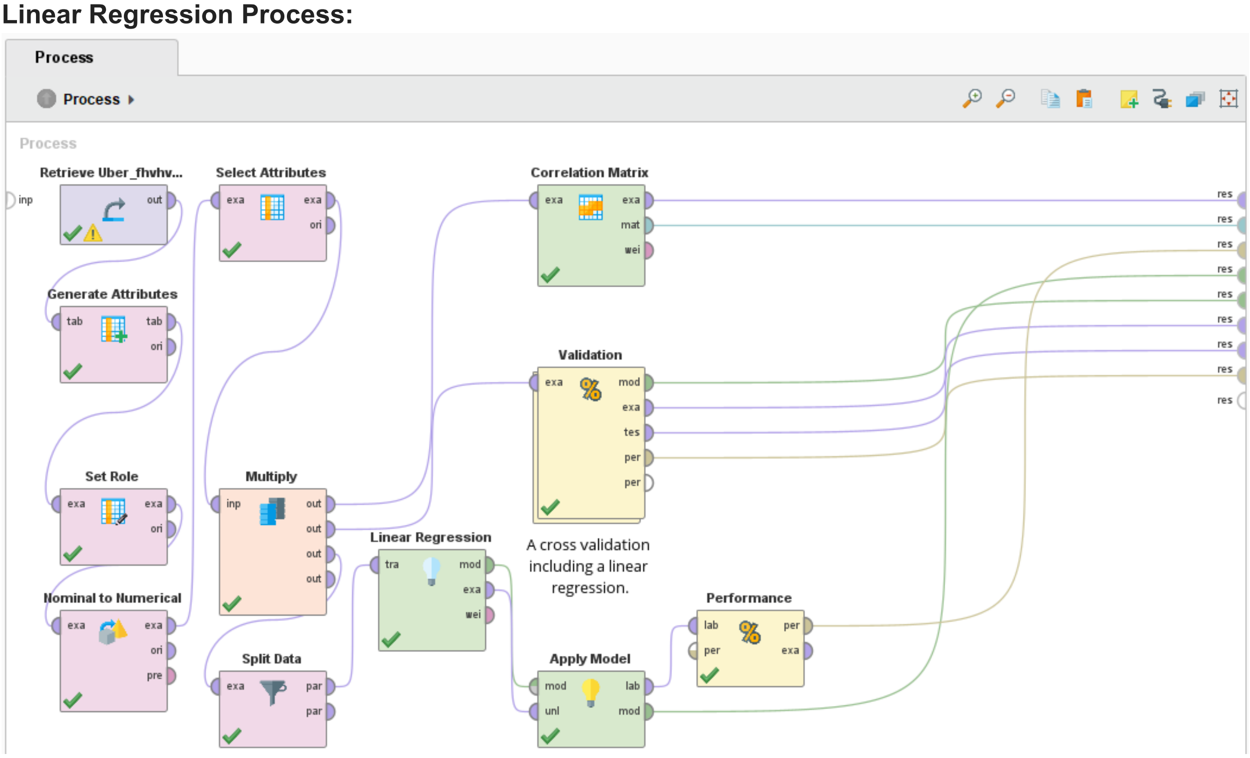
The most beneficial attributes for income predictions are then trip_miles, trip_time, tolls, airport_fee, and cbd_congestion_fee. All linear models produced p-values of 0 for these attributes, illustrating their significance.

Attributes	uber... ↓	driver_...	tips	shared...	shared...	access...	wav_re...	wav_m...	trip_mil...	trip_time	base_p...	tolls	bcf	sales_t...	congest...	airport_...	cbd_co...
uber_pay	1	0.478	0.369	-0.105	-0.071	-0.008	-0.029	-0.105	0.422	0.325	0.773	0.316	0.762	0.555	0.238	0.375	0.319
base_passenger_fare	0.773	0.927	0.423	-0.066	-0.038	0.001	-0.007	-0.066	0.832	0.789	1	0.507	0.998	0.741	0.213	0.454	0.299
bcf	0.762	0.930	0.421	-0.065	-0.038	0.002	-0.004	-0.065	0.837	0.790	0.998	0.509	1	0.740	0.193	0.455	0.276
sales_tax	0.555	0.697	0.321	-0.053	-0.027	0.005	0.002	-0.039	0.577	0.661	0.741	0.133	0.740	1	0.340	0.436	0.258
driver_pay	0.478	1	0.368	-0.029	-0.011	0.006	0.008	-0.030	0.901	0.899	0.927	0.515	0.930	0.697	0.154	0.407	0.225
trip_miles	0.422	0.901	0.320	0.029	0.051	0.004	-0.005	-0.053	1	0.791	0.832	0.547	0.837	0.577	0.033	0.424	0.103
airport_fee	0.375	0.407	0.246	-0.052	-0.040	-0.010	-0.007	-0.032	0.424	0.368	0.454	0.282	0.455	0.436	0.016	1	0.046
tips	0.369	0.368	1	-0.050	-0.038	-0.006	0.001	-0.032	0.320	0.289	0.423	0.223	0.421	0.321	0.130	0.246	0.158
trip_time	0.325	0.899	0.289	0.053	0.095	0.007	0.012	-0.033	0.791	1	0.789	0.428	0.790	0.661	0.170	0.368	0.228
cbd_congestion_fee	0.319	0.225	0.158	-0.053	-0.020	-0.026	-0.011	-0.043	0.103	0.228	0.299	0.205	0.276	0.258	0.845	0.046	1
tolls	0.316	0.515	0.223	-0.032	-0.024	-0.000	-0.009	-0.058	0.547	0.428	0.507	1	0.509	0.133	0.001	0.282	0.205
congestion_surcharge	0.238	0.154	0.130	-0.121	-0.085	-0.026	-0.004	-0.024	0.033	0.170	0.213	0.001	0.193	0.340	1	0.016	0.845
access_a_ride_flag	-0.008	0.006	-0.006	-0.003	-0.003	1	0.036	0.007	0.004	0.007	0.001	-0.000	0.002	0.005	-0.026	-0.010	-0.026
wav_request_flag	-0.029	0.008	0.001	-0.011	-0.008	0.036	1	0.168	-0.005	0.012	-0.007	-0.009	-0.004	0.002	-0.004	-0.007	-0.011
shared_match_flag	-0.071	-0.011	-0.038	0.755	1	-0.003	-0.008	-0.005	0.051	0.095	-0.038	-0.024	-0.038	-0.027	-0.085	-0.040	-0.020
wav_match_flag	-0.105	-0.030	-0.032	-0.001	-0.005	0.007	0.168	1	-0.053	-0.033	-0.066	-0.058	-0.065	-0.039	-0.024	-0.032	-0.043
shared_request_flag	-0.105	-0.029	-0.050	1	0.755	-0.003	-0.011	-0.001	0.029	0.053	-0.066	-0.032	-0.065	-0.053	-0.121	-0.052	-0.053

LINEAR REGRESSION MODELS:

Include linear regression documentation here.

Linear Regression Process:



driver_pay Linear Regression Model from 10-Fold Validation:

driver_pay = 0.118 + 1.617(trip_miles) + 0.010(trip_time) + 0.106(tolls) + 0.283(airport_fee) + 1.66(cbd_congestion_fee)

squared_correlation = 0.909 +/- 0.005
root_mean_squared_error = $5.898 +/- 0.185

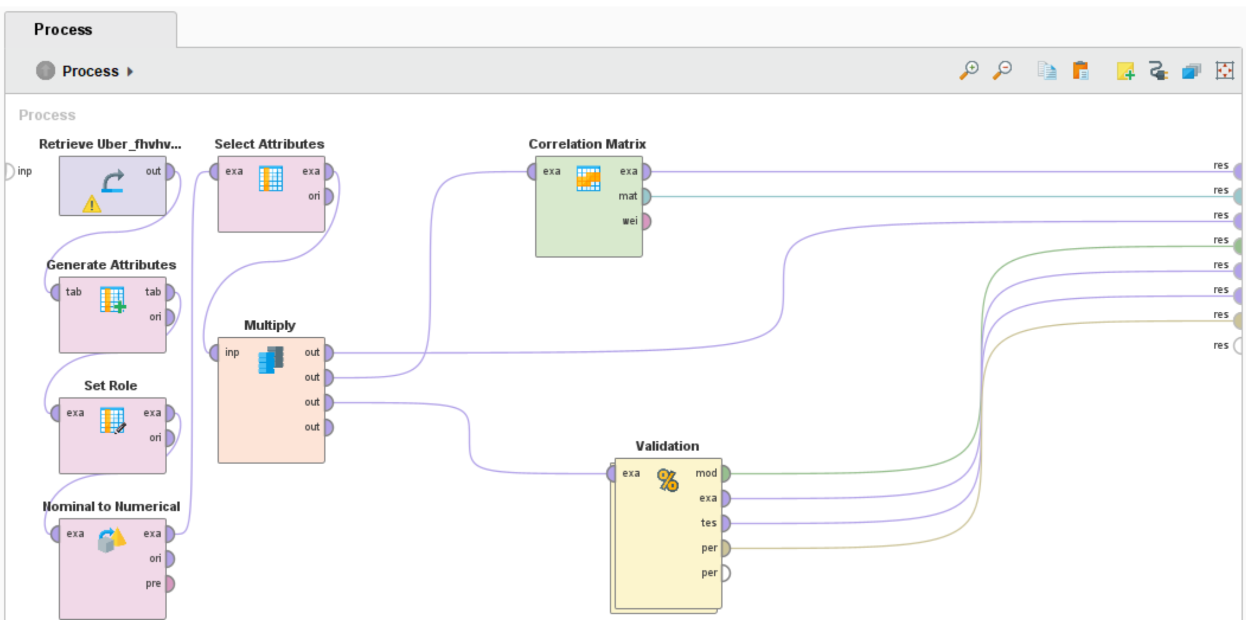
uber_pay Linear Regression Model from 10-Fold Validation:

uber_pay = 0.189 + 0.742(trip_miles) - 0.002(trip_time) + 0.127(tolls) + 3.897(airport_fee) + 4.736(cbd_congestion_fee)
squared_correlation = 0.312 +/- 0.013
root_mean_squared_error = $9.598 +/- 0.173

NEURAL NETWORK MODELS:

Include neural network models here. Hidden layer sizes were established through trial and error, arriving at best performance at 7 nodes in the first layer and 2 nodes in a second layer.

Neural Network Process:



driver_pay Neural Network Model from 10-Fold Validation:

squared_correlation = 0.912 +/- 0.005
root_mean_squared_error = $6.061 +/- 0.502

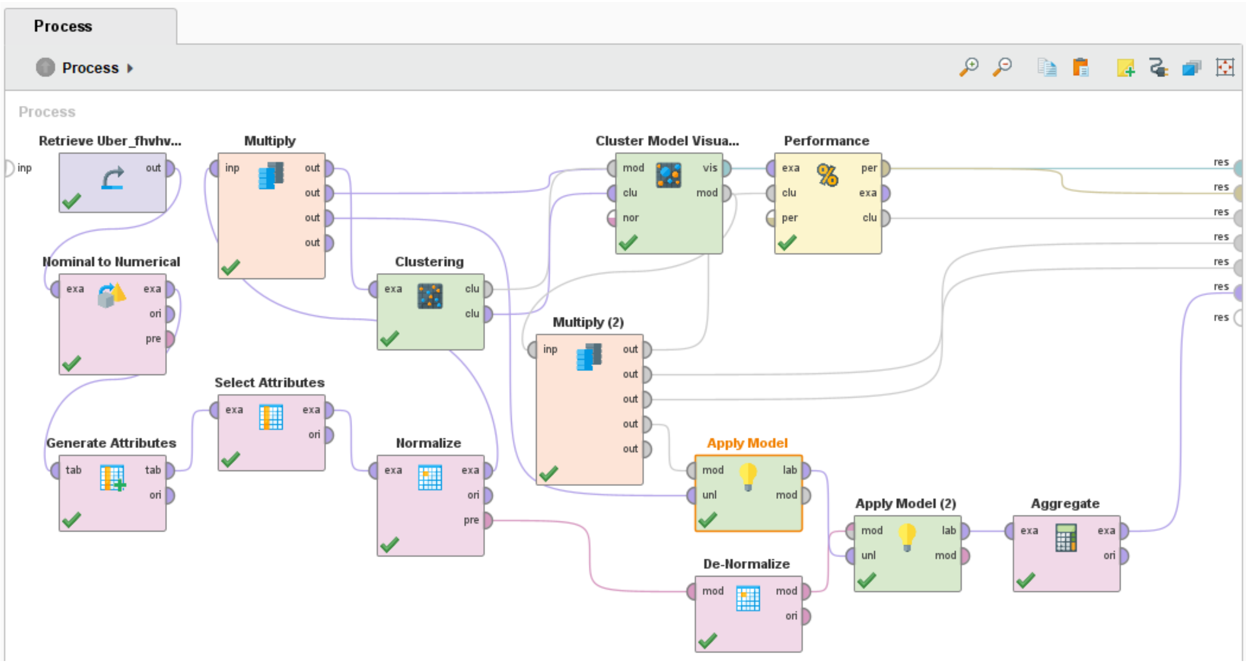
uber_pay Neural Network Model from 10-Fold Validation:

squared_correlation = 0.370 +/- 0.011
root_mean_squared_error = $9.361 +/- 0.401

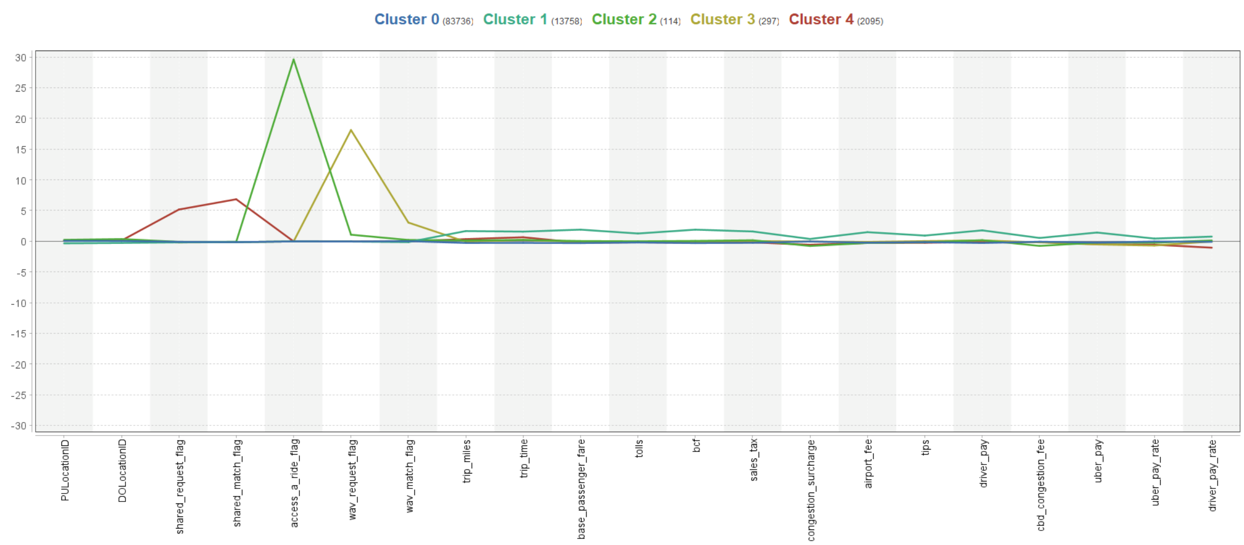
K-MEANS CLUSTERING MODEL:

By applying a K-Means clustering model, distance, trip duration, fare components, tolls, airport fees, shared-ride/wheelchair-accessible/MTA-requested flags, and other operational details were used to group bookings into clusters based purely on mathematical similarity. The number of clusters was optimized through trial and error until the Davies-Bouldin measure closest to zero was achieved. The result was five clusters and a Davies-Bouldin measure of -0.858.

Process:



Centroid Chart:



Centroid Table:

Cluster	PULocat...	DOLoca...	shared_...	shared_...	access...	wav_req...	wav_ma...	trip_miles	trip_time	base_pa...	tolls	bcf	sales_tax	congesti...	airport_...	tips	driver_p...	cbd_con...	uber_pay	uber_pa...	driver_p...
Cluster 0	0.055	0.043	-0.099	-0.146	-0.034	-0.055	0.014	-0.281	-0.274	-0.303	-0.202	-0.303	-0.257	-0.045	-0.235	-0.144	-0.290	-0.082	-0.218	-0.054	-0.097
Cluster 1	-0.344	-0.283	-0.181	-0.146	-0.034	-0.055	-0.146	1.657	1.563	1.886	1.258	1.884	1.588	0.370	1.476	0.917	1.773	0.531	1.416	0.432	0.750
Cluster 2	0.221	0.357	-0.100	-0.085	29.600	1.060	0.203	0.130	0.201	0.030	-0.010	0.051	0.161	-0.774	-0.284	-0.182	0.176	-0.760	-0.227	-0.239	0.123
Cluster 3	0.107	0.094	-0.194	-0.146	-0.034	18.109	3.039	-0.086	0.203	-0.129	-0.160	-0.066	0.029	-0.059	-0.115	0.021	0.132	-0.187	-0.527	-0.698	-0.033
Cluster 4	0.056	0.106	5.161	6.834	-0.034	-0.055	-0.037	0.346	0.648	-0.260	-0.163	-0.258	-0.184	-0.581	-0.273	-0.259	-0.073	-0.139	-0.487	-0.569	-1.051

Cluster Descriptions:

- Cluster 0: Common Rides
- Cluster 1: Long Rides (Often to Airport)
- Cluster 2: MTA Requested Rides
- Cluster 3: Wheelchair-Accessible Rides
- Cluster 4: Shared Rides

Select Cluster Descriptive Statistics:

Uber Pay Descriptive Statistics by Cluster

Cluster	Description	Ride Count	Ride %	Min	Max	Mean	Median	Std Dev	Sum	Uber Pay %
0	Common Rides	83736	83.736%	$ (48.89)	$ 52.11	$ 4.34	$ 3.99	$ 5.92	$ 363,157.84	52.913%
1	Long Rides (Often to Airport)	13758	13.758%	$ (64.02)	$ 196.06	$ 23.25	$ 20.44	$ 20.88	$ 319,893.61	46.609%
2	MTA Requested Rides	114	0.114%	$ (8.98)	$ 17.49	$ 4.24	$ 3.80	$ 5.36	$ 483.15	0.070%
3	Wheelchair-Accessible Rides	297	0.297%	$ (28.55)	$ 59.00	$ 0.77	$ 0.96	$ 7.02	$ 227.93	0.033%
4	Shared Rides	2095	2.095%	$ (49.41)	$ 45.35	$ 1.23	$ 2.28	$ 7.91	$ 2,573.21	0.375%

Uber Pay Rate Descriptive Statistics by Cluster

Cluster	Description	Ride Count	Ride %	Min	Max	Mean	Median	Std Dev
0	Common Rides	83736	83.736%	-0.0970	0.2560	0.0063	0.0052	0.0077
1	Long Rides (Often to Airport)	13758	13.758%	-0.0257	0.2209	0.0102	0.0086	0.0097
2	MTA Requested Rides	114	0.114%	-0.0056	0.0257	0.0048	0.0034	0.0059
3	Wheelchair-Accessible Rides	297	0.297%	-0.0626	0.0581	0.0011	0.0010	0.0069
4	Shared Rides	2095	2.095%	-0.0206	0.0416	0.0021	0.0014	0.0050

Driver Pay Descriptive Statistics by Cluster

Cluster	Description	Ride Count	Ride %	Min	Max	Mean	Median	Std Dev	Sum	Driver Pay %
0	Common Rides	83736	83.736%	$ -	$ 98.85	$ 16.87	$ 14.59	$ 10.09	$ 1,413,011.72	62.627%
1	Long Rides (Often to Airport)	13758	13.758%	$ 2.84	$ 464.99	$ 57.31	$ 51.03	$ 27.09	$ 788,522.28	34.948%
2	MTA Requested Rides	114	0.114%	$ 4.38	$ 75.21	$ 26.01	$ 22.72	$ 15.48	$ 2,964.57	0.131%
3	Wheelchair-Accessible Rides	297	0.297%	$ 0.32	$ 119.68	$ 25.15	$ 19.84	$ 17.42	$ 7,470.08	0.331%
4	Shared Rides	2095	2.095%	$ 2.80	$ 99.16	$ 21.13	$ 18.22	$ 13.29	$ 44,277.64	1.962%

Driver Pay Rate Descriptive Statistics by Cluster

Cluster	Description	Ride Count	Ride %	Min	Max	Mean	Median	Std Dev
0	Common Rides	83736	83.736%	0.0000	0.1479	0.0170	0.0155	0.0053
1	Long Rides (Often to Airport)	13758	13.758%	0.0012	0.3115	0.0223	0.0205	0.0086
2	MTA Requested Rides	114	0.114%	0.0084	0.0402	0.0184	0.0169	0.0050
3	Wheelchair-Accessible Rides	297	0.297%	0.0003	0.0937	0.0174	0.0160	0.0063
4	Shared Rides	2095	2.095%	0.0048	0.0615	0.0111	0.0107	0.0031

INSIGHTS AND RECOMMENDATIONS:

Include insights and recommendations here.

Two more derived attributes were added to the clustering model: uber_pay_rate and driver_pay_rate. These are in US dollars per second. With these attributes, along with uber_pay and driver_pay, descriptive statistics on each cluster can be used to advise Uber on their ride-booking segments. The provided information is very valuable for marketing, further investment, negotiations with drivers, and negotiations with NYC's MTA.

For example, you can see that drivers in all the clusters make a mean pay rate between 0.0111 dollars/sec and 0.0223 dollars/sec. Meanwhile, Uber's portion has a far wider range between 0.0011 dollars/sec (MTA rides) and 0.0102 dollars/sec (long rides). We can also see that 84% of rides (common rides) make up 53% of Uber Pay, while only 14% of rides (long rides) make up almost all of the remaining 47% of Uber Pay. Drivers have a more balanced distribution of their pay with 84% of rides (common rides) accounting for 63% of pay and 14% of rides (long rides) making up 35% of pay.